

Association of fish and n-3 fatty acid intake with the risk of type 2 diabetes: a meta-analysis of prospective studies.

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Results from observational studies on the association of fish and n-3 fatty acid consumption with type 2 diabetes mellitus (T2DM) risk are conflicting. Hence, a meta-analysis was performed to investigate this association from cohort studies. A comprehensive search was then conducted to identify cohort studies on the association of fish and/or n-3 fatty acid intake with T2DM risk. In the highest v. lowest categorical analyses, the fixed or random-effect model was selected based on the homogeneity test among studies. Linear and non-linear dose-response relationships were also assessed by univariate and bivariate random-effect meta-regression with restricted maximum likelihood estimation. In the highest v. lowest categorical analyses, the pooled relative risk (RR) of T2DM for intake of fish and n-3 fatty acid was 1.146 (95 % CI 0.975, 1.346) and 1.076 (95 % CI 0.955, 1.213), respectively. In the linear dose-response relationship, the pooled RR for an increment of one time (about 105 g)/week of fish intake (four times/month) and of 0.1 g/d of n-3 fatty acid intake was 1.042 (95 % CI 1.026, 1.058) and 1.057 (95 % CI 1.042, 1.073), respectively. The significant non-linear dose-response associations of fish and n-3 fatty acid intake with T2DM risk were not observed. The present evidence from observational studies suggests that the intake of both fish and n-3 fatty acids might be weakly positively associated with the T2DM risk. Further studies are needed to confirm these results.